

Machine Learning : KNN regression and Outlier Detection



Question 1 (Regression)

- (i) In the exercise folder you will find a file called regressionExample.csv. Read this file into a NumPy array
- (ii) Use [train_test_split](#) to split the dataset into 20% test and 80% training.
- (iii) Use a [KNeighborsRegressor](#) to build a predictive model for the data and assess its accuracy.

Question 2 (Outlier Detection)

- (i) In the exercise folder you will find a zip file called outlierData.zip. This zip file contains a training file called trainOutlier.csv and a test file called test.csv. This is a regression problem and target value is contained in the last column in each file. Read this data into your program.
- (ii) Build a model using [DecisionTreeRegressor](#) and assess the accuracy.
- (iii) Identify any outliers in the training data using boxplots.
- (iv) Remove the outliers and reassess the new accuracy of the model.
- (v) Try an alternative model such as a [KNeighborsRegressor](#) and record the accuracy.